Junming Liu

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Education

The University of Melbourne, Melbourne, Australia *Setp. 2026 – Present*

Ph.D. in Engineering and IT

- Research Interests: Generative Intelligence, Multimodal Reasoning, Graph Theory

Tongji University, Shanghai, China *Setp. 2023 – Mar. 2026*

M.E. in Computer Science and Technology

- GPA: 90.0/100

Dalian Maritime University, Dalian, China *Sept. 2019 – June 2023*

B.E. in Intelligence Science and Technology

- GPA: 84.0/100

Publications

* Equal Contribution † Corresponding Author

[1] **Junming Liu**, Siyuan Meng, Yanting Gao, *et al.* *Aligning Vision to Language: Annotation-Free Multimodal Knowledge Graph Construction for Enhanced LLMs Reasoning*. ICCV 2025 (Accepted).

[2] **Junming Liu**, Yifei Sun, Weihuang Cheng, *et al.* *ReBrain: Brain MRI Reconstruction from Sparse CT slice via Retrieval-Augmented Diffusion*. WACV 2026 (Accepted).

[3] **Junming Liu**, Yusen Zhang, Rongchao Zhang, *et al.* *Domain-Adaptive Model Merging Across Disconnected Modes*. ICASSP 2026 (Accepted).

[4] Yujin Kang*, **Junming Liu***, Haiyan Cui†. *AI-Driven Assessment of Lip Volume Improvement Using Hyaluronic Acid Fillers: A Comprehensive Analysis*. Aesthetic Plastic Surgery (Accepted).

[5] Aoqi Wu, **Junming Liu**, Yuwei Zhang, *et al.* *AMID: Model-Agnostic Dataset Distillation by Adversarial Mutual Information Minimization*. WWW 2026 (Accepted).

[6] Yanting Gao, Yepeng Liu, **Junming Liu**, *et al.* *Boosting Adversarial Transferability via Commonality-Oriented Gradient Optimization*. PRCV 2025 (Accepted).

[7] Pei Liu, Xin Liu, Ruoyu Yao, **Junming Liu**, *et al.* *HM-RAG: Hierarchical Multi-Agent Multimodal Retrieval Augmented Generation*. ACM MM 2025 (Accepted).

[8] Yuqi Li, Siwei Meng, Chuanguang Yang, Weilun Feng, **Junming Liu**, *et al.* *A Comprehensive Survey of Interaction Techniques in 3D Scene Generation*. IJCAI 2026 (Accepted).

[9] **Junming Liu***, Yifei Sun*, Weihua Cheng, *et al.* *MemVerse: Multimodal Memory for Lifelong Learning Agents*. SIGKDD 2026 (Under Review).

[10] Rongchao Zhang, **Junming Liu**, Haoran Liu, *et al.* *Go With the Fuzziness: Inverse Synthesis of Molecules with Fuzzy Flow Matching*. SIGKDD 2026 (Under Review).

[11] **Junming Liu**, Yifei Sun, *et al.* *Hierarchical Memory Orchestration for Personalized Persistent Agents*. UIST 2026 (Under Review).

[12] Weihua Cheng, **Junming Liu**, Yifei Sun, *et al.* *MGA: Memory-Driven GUI Agent for Observation-Centric Interaction*. ACM MM 2026 (Under Review).

[13] Haodong Lei, **Junming Liu**, Yirong Chen, *et al.* *MemCoT: Test-Time Scaling through Memory-Driven Chain-of-Thought*. ACM MM 2026 (Under Review).

[14] Yuqi Li*, **Junming Liu***, Wenxuan Zeng*, *et al.* *StyleMoD: Style-Controllable Image Generation via Spatio-Temporal Mixture of Diffusion Experts*. ACM MM 2026 (Under Review).

[15] **Junming Liu**, Yuqi Li, *et al.* *Self-Evolving Spatial Reasoning in Vision Language Models via Geometric Logic Consistency*. NeurIPS 2026 (Under Review).

[16] **Junming Liu***, Yuqi Li*, Shiping Wen, *et al.* *Hit-RAG: Learning to Reason with Long Contexts via Preference Alignment*. EMNLP 2026 (Under Review).

Honors and Awards

- **National College Mathematics Competition, National First Prize (1%)** *Dec. 2022*
  Chinese Mathematical Society
- **Excellent Student Scholarship (10%)** *Sept. 2021, 2022*
  Dalian Maritime University
- **Leadership and Communication Scholarship** *Sept. 2020*
  Dalian Maritime University

Research Experience

- **Reasoning in Multimodal Large Language Models** **Shanghai AI Laboratory**
  Advisor: Dr. Ding Wang *Jan. 2025 – Present*
  - **Motivation:** Explore efficient and interpretable reasoning methods for MLLMs.
  - **Methods:** Developed an annotation-free multimodal knowledge graph module to enhance reasoning, designed a long-context prompt comprehension framework to guide outputs and align responses with input semantics, studied partial alignment of different modalities to preserve shared and modality-specific information, and built an agent-memory mechanism for multi-turn interaction.
  - **Results:** Achieved improved performance and robustness across several multimodal benchmarks, with publications in top-tier conferences and ongoing submissions under review.
- **Vision-Language Models for Medical Imaging** **Tongji University**
  Advisor: Prof. Guosun Zeng *Sept. 2023 – Mar. 2026*
  - **Motivation:** Brain medical images contain complex structures that are difficult to interpret reliably, motivating the use of VLMs to enable structured and interpretable reasoning.
  - **Methods:** Used DDPM with ControlNet to perform high-fidelity reconstruction that preserves anatomical consistency across modalities, fine-tuned VLMs for brain abnormality detection and segmentation with traceable and interpretable reasoning, and incorporated privacy-preserving, distributed techniques for efficient lightweight collaboration of large models across heterogeneous hospitals.
  - **Results:** Achieved improved modality-aligned reconstruction and enhanced brain tumor detection and segmentation on BraTS, etc., demonstrating strong robustness even under Non-IID data distributions.
- **Text Sentiment Analysis Based on BERT Model** **Dalian Maritime University**
  Advisor: Prof. Yijia Zhang *Sept. 2021 – May. 2022*
  - **Motivation:** Investigate public opinion tendencies on social media during the COVID-19 pandemic.
  - **Methods:** Enhanced BERT by integrating LDA topic modeling and fine-tuned the combined model for text sentiment analysis.
  - **Results:** Achieved 95% accuracy in analyzing public sentiment on social media posts and submitted a patent titled *"Text Review Sentiment Classification Method Based on Topic-Fused BERT and Medium"*.

Internship Experience

- **Shanghai AI Laboratory** *Jan. 2025 – Present*
  - Conducted research on AI agents.
- **Beijing Jinxixin Network Technology Co., Ltd.** *Oct. 2024 – Dec. 2024*
  - Constructed and trained vertical domain large models for maritime law scenarios.
- **Shanghai NIO Automobile Co., Ltd.** *June. 2023 – Aug. 2023*
  - Controlled steering systems using multimodal sensing for rollover prevention in tire blowout scenarios.

Skills

- Language: Chinese (native), English (IELTS: 7.0), Japanese, Germany
- Programming: C, C++, Python, Java, Go, SQL, Rust, MPI, NCCL, DeepSpeed, DDP, FSDP

Campus Experience

- Member of the Student Union, School of Electronics and Information Engineering, Tongji University
  Sept. 2023 – June. 2024
- Member of the Art Troupe, School of Information Science and Technology, Dalian Maritime University
  Sept. 2019 – June. 2022